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Education

Program	Institute	CGPA/%	Year
Dual Degree (B Tech+M Tech) in Engineering Design	Indian Institute of Technology Madras	8.60/10.0	2026
Class XII (CBSE)	Velammal Bodhi Campus, Chennai	98.2%	2021
Class X (CBSE)	Velammal Bodhi Campus, Chennai	95.8%	2019

Scholastic Achievements

- JEE Advanced 2021: **Top 1.5%**(Among 2 Lakhs); JEE Mains 2021: **99.78%**; National Entrance Screening Test 2021: **Rank 36**
- National Standard Examination in Physics 2021: **State Topper**; International Olympiad of Mathematics 2021: **State rank 25**
- Codeforces - **Specialist**; CodeChef - **3 Stars** (**Global Rank 35** in CodeChef Starters 202); Arithmetic Zetamac - **78 in 2 mins**

Professional Experience

- **Featurely AI** *June 2025 - July 2025 (2 mos)*
  Applied AI Intern San Francisco Bay Area Based
  - Featurely is a US AI startup backed by **Bling Capital**, whose partners have backed companies such as **Palantir** and **Lyft**.
  - Built an explainable **human-behavior simulation model** under data constraints using **LLMs**, **Probablistic Modelling**, **graph-based memory system** simulating user behavior, for faster, user-focused product development and feature prioritization.
  - Engineered **episodic memory system** (Neo4j, Pinecone) with recursive retrieval (**vector RAG and graph RAG**).
  - Deployed the system (**FastAPI** + **Docker** on **AWS EC2** backend; **Vercel** frontend), enabling remote testing by stakeholders.

- **Caterpillar** *Jan 2025 - May 2025 (5 mos)*
  Machine Learning Intern, Global Data Science Team (Optimisation Pod) Chennai, India
  - Developed **Multi-Agent Reinforcement Learning** dispatch system for mining fleets using **Proximal Policy Optimisation** algo.
  - Engineered **multi-objective Reward Framework**: Productivity **8-10%**↑, truck congestions **30%**↓, power efficiency **10%**↑.
  - Automated mine layout generation, cutting setup time from **hours to minutes** for rapid training (across 50+ site layouts).

Research Experience

- **ADOPT Lab, IIT Madras**
  - (**Patent** Applied) Proposed and evaluated a **topology-preserving synthetic tabular data generation** approach for scarce and non-convex datasets, demonstrating improved stability, downstream task utility and fidelity over standard baselines.

Projects

- **Fraud Ring Investigator Arena: RL Benchmark for Sequential Anti Money Laundering Investigation**
  - Built an **OpenEnv benchmark** for fraud triage with observability, action costs, delayed effects, and clear/escalate decisions.
  - Designed typed investigation/intervention actions with deterministic baselines on **easy/medium/hard** tracks for **RL agents**.

- **End-to-End RL Fine-Tuning Pipeline for DeepSeek-7B using Custom Reward Modeling**
  - Engineered a **Reinforcement Learning (GRPO) pipeline** to **fine-tune** a 7B LLM, implementing a custom reward function.
  - Optimized training efficiency using **gradient checkpointing** and **8-bit quantization** within the Hugging Face trl ecosystem.

- **Efficient LLM Adaptation: QLoRA-based Supervised Fine-Tuning of DeepSeek-7B**
  - Implemented **SFT**[2] using **LoRA**[3] adapters and 4-bit quantization (**QLoRA**[4]) to specialize LLMs for mathematical reasoning.
  - Integrated **gradient accumulation and mixed-precision training** to enable **efficient large-model tuning on consumer-grade GPUs**.

- **Productionizing LLMs: Quantization and Performance Profiling for Efficient Inference**
  - Executed a model optimization pipeline reducing **memory footprint by 70%** via 4-bit NF4 and Int8 **quantization** techniques.
  - Benchmarked **inference latency and throughput** across precision formats to identify optimal configurations for **edge deployment**.

- **Implementation of Custom Deep Learning Architectures for Computer Vision**
  - Implemented **YOLOv2 for Car Detection**, using IoU[5]/NMS[6] for real-time bounding boxes in **autonomous driving**.
  - Designed **U-Net for Semantic Segmentation**, achieving 80% pixel accuracy on **object outlining tasks** (noisy data).

- **Clinically RAG: Chat-First Healthcare AI Research Assistant**
  - Developed a **chat-first healthcare assistant** using **Next.js** with a **7-stage LangGraph** pipeline on a 21-article corpus.
  - Implemented **hybrid retrieval**, citation verification, and critic-based repair for **corpus-grounded** answers.

- **Foundational Machine Learning and Statistical Modeling**

  Supervised Learning Models
  - Implemented regression (**Ridge, Lasso**), on high-dim 2D data, achieving 99.5% MSE reduction and 86% feature sparsity.
  - Derived **Bayesian LDA/QDA**[7] **classifiers** with asymmetric loss on 2D (binary/multiclass) data, reducing errors by 72%.

  Unsupervised Learning Models
  - Implemented **clustering (DBSCAN, spectral)** for non-convex data, achieving 100% visual purity via Laplacians.
  - Developed **Expectation Maximization** for **Gaussian Mixture Models** (ELBO-monitored cluster alignment) on 2D data.

Skills

- **Programming:** C++, Python, SQL; **Tools:** Git, Docker, Redis, AWS, AirFlow,FastAPI, Neo4j, Pinecone
- **Utility Libraries:** statsmodels, PyTorch, Tensorflow, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Keras, SciPy

Extracurricular

Tech Contingent in **Inter IIT Tech Meet 13.0** (**Represented IITM** competing against **23 IITs**; Organised by **ISRO**):
- Generated a lunar map at sub-pixel resolution by analyzing X-ray fluorescence data from Chandrayaan-2's CLASS instrument.

Relevant Courseworks

- Probability, Statistics and Stochastic Processes; Linear Algebra; Differential Equations; Series and Matrices
- Machine Learning and Deep Learning Specialisation*; Pattern Recognition and Machine Learning
- Fine-tuning & RL for LLMs: Intro to Post-training**; First Principles Tools; Data Analysis & Visualization in R/Python/SQL
- Operations Research; Operations Management; Principles of Economics; Econometrics; Business Analytics

Positions of Responsibility

- **Core Team Member**, Wellness Team, IIT Madras *Aug 2022 - May 2025 (3 yrs)*
  Managed a team of **50** coordinators; Led numerous wellness events; Offered peer counselling support to students.
- **Department Coordinator**, Academic Mentorship Cell, IIT Madras *Sep 2023 - May 2024 (1 yr)*
  Facilitated academic journeys for **20** mentees by matching them with suitable mentors and monitoring mentees' progress.

*- Stanford Online, **-DeepLearning.Ai, 1-Root Mean Squared Error, 2-Supervised Fine Tuning, 3-Low-Rank Adaptation, 4-Quantized Low-Rank Adaptation, 5-Intersection Over Union, 6-Non-Max Suppression, 7-Linear Discriminant Analysis/Quadratic Discriminant Analysis